

JOHN SWAROOP

Founder · Full-Stack Engineer · Applied AI

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PROFILE

Full-stack engineer and founder with 5+ years taking products from early ideas to production. Works across TypeScript, Python, React/Next.js, Node.js, data and infrastructure, with production experience in agentic workflows, RAG, document intelligence and realtime voice. Comfortable in zero-to-one environments that require product judgment, system architecture and hands-on execution.

CORE STRENGTHS

ZERO-TO-ONE PRODUCT	FULL-STACK ENGINEERING	APPLIED AI SYSTEMS
Product strategy · architecture · prototyping · technical leadership · launch	TypeScript · Python · React/Next.js · Node.js · APIs · data · infrastructure	Agents · RAG · voice · document intelligence · tool calling · model routing

PROFESSIONAL EXPERIENCE

Founder & CTO · DcernX

2025 — Present

- Building an AI operating system for venture capital, private equity and investment teams in partnership with a UK-based accelerator, centered on agentic research, document intelligence and collaborative diligence.
- Designed multi-agent research workflows that coordinate web search, uploaded documents, specialized agents, tool calls and structured outputs through stateful orchestration.
- Built evidence-backed analysis with citation grounding, claim-to-evidence mapping, risk extraction and multi-document reasoning.
- Developed ingestion and retrieval pipelines using Docling, PageIndex, RAG and hierarchical/vector retrieval strategies.
- Own product and technical architecture across the Next.js/TypeScript application, Python AI workflows, data infrastructure, model/provider routing and deployment.

Founder / Product & Engineering Lead · P101 Labs

2024 — Present

- Built and shipped AI and full-stack products for startups and SMEs across voice, chat, interview automation, document intelligence and workflow automation.
- Partnered directly with founders to translate ambiguous operational problems into product scope, architecture and production-ready software.
- Owned interfaces, APIs, data models, integrations, streaming and asynchronous workflows alongside the AI systems embedded in each product.

Independent Consultant — Full-Stack & AI · Selected engagements

2023 — Present · Concurrent

- Partnered with startups and SMEs on product architecture, rapid prototyping and production engineering across operations, customer support and document intelligence.
- Contributed across product definition, frontend, backend, integrations and deployment based on the needs of each engagement.

SELECTED WORK & TECHNICAL DEPTH

A closer look at selected product work and the systems behind it.

SELECTED PRODUCT WORK

INTERACTIVE DESIGN PLATFORM 3D Garden & Home Designer A visual product that helps users create polished 3D garden and home mockups through an approachable design workflow. Worked across the interactive product experience and supporting platform, helping translate specialist 3D-design concepts into a tool for everyday users.	GOVTECH · WORKFLOW AUTOMATION Krooki Management System A management platform supporting the Government of Oman in digitizing construction application and approval processes. Helped turn a multi-stage operational process into a structured digital workflow for application management, reviews, approvals and stakeholder coordination.
APPLIED AI · REALTIME VOICE Interview Practice & Screening Platform An AI-powered voice platform for realistic interview practice and structured candidate screening. Built end-to-end across question generation, live voice conversations, session management, evaluation and actionable feedback.	DEALER ERP · COMMERCE PLATFORM Used-Car Dealer ERP An end-to-end operating system combining vehicle inventory, lead management and a customizable customer-facing storefront. Built automated storefront generation, lead capture and pipelines, dealer branding and merchandising controls across the full stack.

EARLIER EXPERIENCE

Founding Full-Stack Engineer · CultureCo Sep 2024 — Dec 2024 - Built the full-stack creator commerce platform enabling creators to sell products, manage communities and grow their businesses. - Contributed across product development, frontend, backend and integrations in an early-stage, zero-to-one environment.	
Founding Engineer · Truts Dec 2021 — Mar 2024 - Built a Web3 discovery and quest platform from the ground up across frontend, backend, blockchain integrations and product development. - Owned broad engineering scope as an early team member, taking product capabilities from concept through production release.	

SELECTED APPLIED AI SYSTEMS

Deep research & diligence Multi-agent research across documents, web sources and structured data, producing grounded findings, citations, claim/evidence links and risk analysis.	Realtime voice agents Streaming STT → LLM → TTS systems with VAD, tool execution, LiveKit Agents, Whisper and production integration workflows.
Workflow agents & automation Knowledge-grounded, tool-enabled agents with multi-turn state, configurable workflows, external actions and human review.	Document intelligence pipelines Document ingestion, extraction, hierarchy-aware indexing, retrieval and structured analysis across complex files.

TECHNICAL STACK

PRODUCT & FRONTEND	TypeScript · React · Next.js · Tailwind · React Query · Zustand · interaction design
BACKEND & REALTIME	Python · Node.js · Express · REST APIs · WebSockets/SSE · async workflows · system design
APPLIED AI	LangGraph · Vercel AI SDK · LangChain · LiveKit Agents · RAG · tool calling · structured outputs
MODELS	OpenAI · Claude · Gemini · DeepSeek · Qwen · Fireworks AI · open-weight inference · provider routing
DATA & INFRASTRUCTURE	PostgreSQL · MongoDB · Convex · Supabase · Redis · AWS · Docker · Cloudflare · Railway · Coolify · Linux